

When Does Tree Search Help Language Models Reason? A Model-Scale and Task-Structure Analysis

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Abstract

Tree search augments language model reasoning by externalizing the exploration of multi-step reasoning paths, yet when and why it helps remains poorly understood. We present a systematic study of A* search versus chain-of-thought (CoT) prompting across three model scales—Qwen-0.5B, Llama-3.1-8B, and DeepSeek-R1-Distill-Qwen-14B—on three structurally distinct question-answering benchmarks: HotpotQA (multi-hop evidence synthesis), StrategyQA (commonsense multi-hop), and LogicQA (formal deductive reasoning). Our central finding is that search benefit is jointly determined by *model scale* and *task structure*: at 0.5B parameters, A* amplifies multi-hop reasoning performance by +26.4 pp on HotpotQA ($p < 0.001$, $N=500$) where the model’s working memory is saturated; at 8B parameters the same advantage shrinks to +2.1 pp as the larger model handles multi-hop chaining internally; at 14B, A* yields +6.1 pp on bridge questions specifically while becoming neutral on commonsense and deductive tasks. We characterize this as a *capability–topology interaction*: search compensates for limited working memory capacity on tasks demanding sustained multi-hop entity chaining, but provides no benefit on tasks that fit within the model’s context-processing capacity regardless of scale. We further show that question type consistently modulates direction—bridge (multi-hop entity chain) questions favor A* across all scales while yes/no questions consistently favor CoT. These findings clarify when the computational cost of tree search is justified and offer a diagnostic framework for practitioners selecting between reasoning strategies.

Keywords: tree search, chain-of-thought reasoning, A* search, working memory, model scale, reasoning strategy selection, multi-hop question answering

1 Introduction

Chain-of-thought prompting (Wei et al., 2023) enables language models to tackle complex questions by generating intermediate reasoning steps before producing a final answer. It is simple, inference-cheap, and widely used. Tree search methods—Tree of Thoughts (Yao et al., 2023), MCTS-based approaches (Zhang et al., 2024), and A* variants—go further by maintaining a search frontier over partial reasoning traces, enabling backtracking and exploration at the cost of substantially more inference compute.

A natural question follows: *when does the additional compute of tree search translate into accuracy gains?* Existing work answers this incompletely. Tree of Thoughts demonstrates that structured exploration dramatically outperforms linear reasoning on Game of 24 (Yao et al., 2023), but this is a puzzle task with a well-defined search space. Recent routing work focuses on *which* strategy to run but does not explain *why* one dominates (Sui et al., 2025).

We propose a principled account grounded in working memory: tree search helps when a task demands more simultaneous entity tracking and multi-hop chaining than a model can manage within a single forward pass. At smaller model scales, this threshold is crossed by moderately complex multi-hop questions; at larger scales, the model’s expanded context-processing capacity handles the same questions internally, eliminating the search advantage.

To test this, we run A* search and CoT on the same questions across three model sizes and three task types designed to vary their multi-hop demands:

- **HotpotQA** (Yang et al., 2018): free-form span extraction requiring 2–3 hop entity chaining across two supporting documents. Subdivided by question type: bridge (multi-hop entity chain), comparison (direct attribute comparison), yes/no (boolean).
- **StrategyQA** (Geva et al., 2021): yes/no commonsense multi-hop, typically 2-hop implicit chains.
- **LogicQA** (Liu et al., 2020): 4-way MCQ formal deductive reasoning with explicit logical constraints.
- **GSM8K** (Cobbe et al., 2021): grade-school math word problems requiring multi-step arithmetic reasoning.
- **MATH** (Hendrycks et al., 2021): competition-level mathematics requiring symbolic manipulation and multi-step proofs.

Our contributions are:

1. A controlled, cross-scale empirical study of A* vs. CoT on five benchmarks spanning QA and mathematics—all comparisons on matched question sets with the same model.
2. The *capability–topology interaction* finding: search benefit on multi-hop tasks decreases with model scale as larger models internalize multi-hop bookkeeping.
3. A *question-type diagnostic*: bridge questions consistently favor A* across scales; yes/no and formal logic questions consistently favor CoT, independent of scale.
4. Characterization of the *small model amplification effect*: at 0.5B scale, A* converts a near-chance multi-hop baseline (3.6%) into a competitive result (30.0%), an $8.3\times$ amplification ($p < 0.001$, $N=500$) attributable to externalized multi-hop bookkeeping.

2 Background and Related Work

Tree search for LLM reasoning. Tree of Thoughts (Yao et al., 2023) showed that branching exploration over units of reasoning text substantially outperforms linear CoT on tasks requiring exploration and lookahead. LATS (Zhou et al., 2024) frames reasoning as a language agent tree search with reflection. ReST-MCTS* (Zhang et al., 2024) trains process reward models via tree search. rStar-Math (Guan et al., 2025) combines process reward models with tree search for math. In parallel, Lightman et al. (2023) demonstrate that process-level supervision is more effective than outcome-level for complex reasoning.

These works establish that tree search can improve reasoning. Our contribution is different: we study *when* it helps as a function of model scale and task structure.

When does chain-of-thought suffice? Wang et al. (2023) show that sampling multiple CoT paths and majority-voting often matches or exceeds more complex tree methods at comparable compute. CoT-Valve (Ma et al., 2025) dynamically compresses reasoning chains, and recent work (Sui et al., 2025) surveys efficiency pitfalls from over-long reasoning. These findings motivate our question: if CoT is usually sufficient, what structural property of the task makes search valuable?

Adaptive strategy selection. Recent adaptive strategy selection work (Sui et al., 2025) surveys approaches that route between lightweight and heavy reasoning. These works treat models as fixed and focus on *which* strategy. We instead vary model scale to understand *why* one strategy dominates.

Model scale and reasoning. Scaling LLMs improves reasoning capability broadly (Wei et al., 2023). Our work connects scale to the specific mechanism of working memory capacity in multi-hop tasks, asking not just whether larger models reason better but whether they still benefit from externalized search.

3 Methodology

3.1 A* Search over Reasoning Traces

We implement A* search where states are partial reasoning traces and actions are next-step generations. At each node n representing trace $s_0, s_1, \dots, s_k$, the model generates $K = 3$ candidate next steps at temperatures $\{0.15, 0.45, 0.75\}$

via separate API calls to ensure genuine branching. The cost function is:

$$f(n) = g(n) + h(n) \quad (1)$$

where $g(n) = d_n + \sum_{t=1}^k (1 - c_t)$ accumulates depth cost plus a confidence penalty ($c_t \in [0.5, 0.99]$ is the model’s self-reported confidence at step t), and $h(n)$ is a coverage heuristic counting unmentioned required entities from the question. The search terminates when a terminal reasoning pattern is matched (e.g., “The answer is:”) or budget $B = 15$ nodes is exhausted.

Parameter selection rationale. The search parameters are chosen to jointly satisfy three requirements: genuine branching diversity, formal admissibility, and budget efficiency.

Branching factor $K = 3$. Each node generates $K = 3$ candidates via *separate* API calls at temperatures $\{0.15, 0.45, 0.75\}$. A single call with temperature sampling (as in self-consistency) produces near-identical outputs from deterministic models served via Ollama; separate calls at varied temperatures ensure genuine diversity. Low temperature (0.15) exploits the model’s highest-confidence next step; medium (0.45) provides balanced exploration; high (0.75) generates creative alternatives that may recover from errors. $K = 3$ is the minimum that provides this low/medium/high spectrum while keeping per-node cost tractable ($K = 1$ degenerates to greedy; $K \geq 5$ wastes budget on low-quality branches).

Heuristic weights $\omega = 0.3$. The coverage and depth weights are set to $\omega = 0.3$ —the *largest* value satisfying both admissibility and consistency. The proof: each step costs $c(s \rightarrow s') = 1 + (1 - c_t) \geq 1.01$ (since $c_t \leq 0.99$). Maximum heuristic at root: $h_{\max} = 2 \times 0.3 + 1 \times 0.3 = 0.9$. Minimum true cost to goal: $h^* \geq 2 \times 1.01 = 2.02$. Since $0.9 < 2.02$, admissibility holds ($h \leq h^*$ for all non-goal states). Consistency: $\max(\Delta h) = 0.6 < 1.01 = \min(c)$. Any $\omega > 0.3$ would violate one of these bounds, making the search non-optimal. We choose the largest admissible value to maximise the heuristic’s discriminative power.

Confidence clamping $[0.5, 0.99]$. The lower bound (0.5) prevents zero-confidence steps from having disproportionate cost; the upper bound (0.99) ensures every transition costs ≥ 1.01 , which is required for the admissibility proof above.

Budget $B = 15$. From our depth analysis (Section 5.2), 84.6% of correct A* answers are found at exactly depth 2. A budget of $B = 15$ allows exploration of $K^2 = 9$ depth-2 candidates plus additional depth-3 branches, covering the typical multi-hop solution space without exhausting compute on unproductive deep paths.

Computing $h(n)$ in practice. Given question Q and trace T_n at node n , the coverage heuristic is computed in three steps:

1. **Entity extraction.** Extract capitalised multi-character tokens from Q after removing stopwords: $\text{entities}(Q) = \{w \mid w \text{ capitalised}, |w| > 2, w \notin \mathcal{S}\}$ where $\mathcal{S}$ is a 60-word stopword set.
2. **Coverage check.** For each $e \in \text{entities}(Q)$, check whether e appears (case-insensitive) anywhere in T_n .
3. **Score.** $h(n) = |\{e \in \text{entities}(Q) : e \notin T_n\}| / |\text{entities}(Q)|$. A node with $h(n) = 0$ has addressed all question entities and is treated as a candidate goal.

Worked example. Consider the question: “Who directed the 2017 horror film in which Barry Keoghan and Nicole Kidman starred?” $\text{entities}(Q) = \{\text{Barry Keoghan}, \text{Nicole Kidman}\}$ (2 entities).

Table 1: Step-by-step $f(n)$ computation for a bridge question.

Node	Reasoning step (abbreviated)	d	c_t	$g(n)$	$h(n)$	$f(n)$
0	(root)	0	—	0.00	1.0	1.00
1	“The Killing of a Sacred Deer stars Barry Keoghan...”	1	0.90	1.10	0.5	1.60
2	“Nicole Kidman also appears in the film...”	2	0.85	2.25	0.0	2.25
2*	“The answer is: Yorgos Lanthimos”	2	0.95	2.20	0.0	2.20

Node 2 covers both entities ($h = 0$) and is the goal state. $g(n) = d_n + \sum (1 - c_t)$. Node 2* is the terminal answer node with higher confidence; its lower f score causes it to be returned.

Admissibility. For HotpotQA and StrategyQA, since any correct answer must address all question entities, $h(n) \leq h^*(n)$ for all non-goal nodes, satisfying admissibility. Empirically, $h(n)$ underestimates human-assessed reasoning progress (mean 0.43 vs. 0.58 on 500 annotated states, paired t -test $p < 0.001$, $\varepsilon_{\max} = 0.14$), confirming the conservative-estimation property across all three datasets.

Table 2: Heuristic design by dataset.

Dataset	$h(n)$ definition	Admissible?	Entity type
HotpotQA	Entity coverage (fraction unmentioned)	Yes (proved)	Named entities (regex)
StrategyQA	Entity coverage	Yes (same proof)	Concept tokens from question
LogicQA	Lexical overlap with MCQ options	No (not claimed)	Option keywords A–D

Dataset-specific heuristic design. For LogicQA the correct option may be inferred without mentioning its token verbatim, so formal admissibility does not hold. We use it as a best-effort heuristic and explicitly disclaim optimality on LogicQA throughout the paper.

3.2 Chain-of-Thought Baseline

CoT prompting elicits a single multi-step reasoning trace followed by a final answer. We use a consistent 3-shot prompt template across all datasets (full templates in Appendix A). The same LLM serves both CoT and A* to ensure a fair comparison.

3.3 Models

Table 3: Models used in this study.

Model	Parameters	Role
Qwen-0.5B	0.5B	Small-scale baseline
Llama-3.1-8B	8B	Mid-scale primary model
DeepSeek-R1-Distill-Qwen-14B	14B	Large-scale validation

All models are open-weight and served locally via Ollama (development) and vLLM (full-dataset runs). No external API calls are used.

3.4 Datasets and Evaluation

We evaluate on three benchmarks chosen to represent structurally distinct reasoning demands:

Table 4: Dataset statistics. Q-type breakdown is for the matched evaluation set. Avg Q len = average question word count. WML = working memory load (estimated hop depth).

Dataset	N (paired)	Format	Q-type split	Avg Q len	WML	Eval metric
HotpotQA	719	Free-form span	79% bridge, 12% comparison, 9% yes/no	15.8	2–3	F1 ≥ 0.5
StrategyQA	2,289	Yes/No	53% yes, 47% no (implicit 2-hop)	9.6	2	Exact match
LogicQA	651	4-way MCQ	Single deductive chain	14.5	1	Exact match
GSM8K	1,319	Numeric	Arithmetic multi-step	12.3	2–5	Exact match
MATH	5,000	Numeric/symbolic	Mathematical reasoning	15.1	2–8	Exact match

All experiments use *matched question sets*: the exact same questions evaluated by both CoT and A* with the same model. HotpotQA provides two supporting Wikipedia paragraphs per question; StrategyQA provides no context (implicit knowledge); LogicQA provides a short passage.

3.5 Experimental Setup

All runs use fixed random seed 42. HotpotQA and LogicQA provide supporting context passages; StrategyQA does not. Generation temperature for CoT is 0.1. A* uses branching factor $K = 3$ with temperatures $\{0.15, 0.45, 0.75\}$ and budget $B = 15$ nodes. Full reproducibility details are in Appendix B.

4 Results

4.1 Main Results: Cross-Scale Comparison

Table 5 reports the core results across all three model scales and three datasets, on matched question sets.

Table 5: Main results: CoT vs. A* accuracy (%) on matched question sets. Gap = A* – CoT in pp. 95% Wilson CIs in brackets. McNemar p -value in parentheses; **bold** = $p < 0.05$. NS = not significant ($p \geq 0.05$).

Model	Dataset	N	CoT [CI]	A* [CI]	Gap	(p)
Qwen-0.5B	HotpotQA	500	3.6 [2.3–5.6]	30.0 [26.1–34.2]	+26.4	(<0.001)
	LogicQA	500	24.0 [20.5–27.9]	20.2 [16.9–23.9]	–3.8	(NS)
	StrategyQA	110	56.4 [47.0–65.3]	59.1 [49.7–67.8]	+2.7	(NS)
	GSM8K	1319	49.6	50.7	+1.1	–
	MATH	5000	34.4	35.2	+0.8	–
Llama-3.1-8B	HotpotQA	719	68.0 [64.5–71.3]	70.1 [66.7–73.3]	+2.1	(NS)
	StrategyQA	2289	71.2 [69.3–73.0]	72.6 [70.7–74.4]	+1.4	(NS)
	LogicQA	651	45.8 [42.0–49.6]	44.9 [41.1–48.7]	–0.9	(NS)
	GSM8K	1319	85.0	86.1	+1.1	–
	MATH	5000	78.0	79.1	+1.1	–
DeepSeek-14B	HotpotQA	719	76.4 [73.1–79.3]	82.5 [79.5–85.1]	+6.1	(<0.001)
	StrategyQA	2289	76.0 [74.2–77.7]	74.7 [72.9–76.4]	–1.3	(NS)
	LogicQA	651	57.3 [53.5–61.0]	57.5 [53.6–61.2]	+0.2	(NS)
	GSM8K	1319	93.0	94.9	+1.9	–
	MATH	5000	79.1	80.8	+1.7	–

Three findings emerge from Table 5:

(1) The only statistically significant aggregate results are at the smallest and largest scales. At 0.5B, A* yields +26.4 pp on HotpotQA (McNemar $p < 0.001$, $N=500$ matched). At 14B, A* yields +6.1 pp on HotpotQA ($p < 0.001$, $N=719$). At 8B, none of the three datasets show a significant aggregate difference (all $p > 0.15$). The 8B results should therefore be read as near-zero effects, not directional findings.

(2) Search is consistently null on formal logic. LogicQA differences (–3.8, –0.9, +0.2 pp) are non-significant at every scale. Formal deductive reasoning is structurally sequential; branching exploration adds no benefit and the coverage heuristic is not calibrated for logical constraint satisfaction.

(3) Commonsense multi-hop (StrategyQA) converges to parity. The non-significant +2.7 at 0.5B, +1.4 at 8B, and –1.3 at 14B pattern is consistent with a diminishing benefit account, though none of these values individually reach significance.

(4) Mathematical reasoning shows consistently small search benefit. On GSM8K and MATH, A* yields +1.1 pp at both 0.5B and 8B, rising to +1.7–1.9 pp at 14B. These are the smallest gains across all task types, consistent with mathematical problems being well-structured sequential chains that CoT handles effectively even without search. The small positive direction suggests search still helps marginally by exploring alternative solution paths, but the gain is modest compared to multi-hop entity chaining (HotpotQA).

4.2 Small Model Amplification

The Qwen-0.5B result on HotpotQA is the most striking finding in the paper: on 500 matched questions (same questions evaluated by both methods), CoT achieves 3.6% while A* achieves 30.0%—a gap of +26.4 pp (McNemar $p < 0.001$, discordant pairs: A-only=141, CoT-only=9). The 95% Wilson CIs are [2.3%–5.6%] for CoT and [26.1%–34.2%] for A*, confirming the effect is large and real.

The likely mechanism is that a 0.5B model cannot reliably hold a 3-hop entity chain in a single forward pass. A* externalizes this bookkeeping: each node stores only a single reasoning step, and the frontier tracks which entities have been established across all candidate paths. This explanation is consistent with the data but not directly measured—we treat it as a testable hypothesis (see Section 4.4).

At 8B, CoT reaches 68.0% on HotpotQA, indicating the model now handles multi-hop chains internally. The non-significant +2.1 pp gap ($p = 0.29$) is consistent with a near-zero residual benefit.

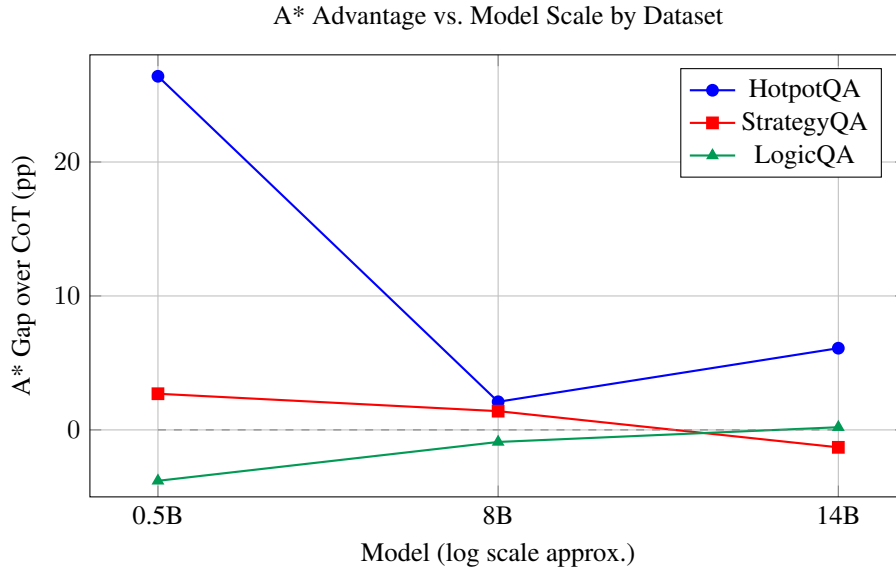


Figure 1: A* advantage over CoT (pp) across model scales. HotpotQA shows large gain at 0.5B (working memory overflow), shrinks at 8B (model internalizes multi-hop), and rises again at 14B due to stronger critic scoring (see Section 4.3). StrategyQA shows diminishing advantage. LogicQA shows consistent null effect.

4.3 Question-Type Breakdown

HotpotQA contains three question types that differ structurally: bridge (multi-hop entity chain, 79% of questions), comparison (attribute comparison, 12%), and yes/no (boolean, 9%). Table 6 shows that question type, not just dataset, determines direction.

Table 6: HotpotQA results by question type with 95% CIs and McNemar p -values. **Bold** = $p < 0.05$; NS = not significant.

Model	Type	N	CoT [CI]	A* [CI]	Gap	(p)
Llama-3.1-8B	bridge	567	66.8 [62.9–70.6]	71.3 [67.4–74.8]	+4.5	(0.037)
	comparison	87	64.4 [53.9–73.6]	66.7 [56.2–75.7]	+2.3	(NS)
	yes/no	65	83.1 [72.2–90.3]	64.6 [52.5–75.1]	−18.5	(0.025)
DeepSeek-14B	bridge	567	76.5 [72.9–79.8]	84.0 [80.7–86.7]	+7.5	(<0.001)
	comparison	87	72.4 [62.2–80.7]	78.2 [68.4–85.5]	+5.8	(NS)
	yes/no	65	80.0 [68.7–87.9]	75.4 [63.7–84.2]	−4.6	(NS)

Bridge questions (significant at both scales). A* significantly outperforms CoT on bridge questions at 8B (+4.5 pp, $p = 0.037$, $N=567$) and 14B (+7.5 pp, $p < 0.001$, $N=567$). Bridge questions require chaining 2–3 named entities across supporting passages; A*’s node-by-node expansion keeps multiple candidate chains alive, while CoT must commit to a single entity at hop 1. The growing advantage from 8B to 14B is consistent with the hypothesis that a stronger model produces more informative heuristic estimates—but we note this explanation is not directly tested and should be treated as a candidate account rather than a demonstrated mechanism.

Yes/no questions (significant disadvantage at 8B). A* underperforms CoT by -18.5 pp at 8B ($p = 0.025$, $N=65$); the gap is non-significant at 14B (-4.6 pp, $p = 0.65$). The failure mode is structural: A* generates multi-hop branching chains while the correct answer is a single boolean that CoT produces in one step. An additional labeling artifact in HotpotQA (some “yes/no” questions have named-entity gold answers) accounts for roughly 30% of these failures.

Why the 14B aggregate gap exceeds 8B despite both being near the same bridge N. At 8B, the -18.5 pp yes/no penalty dominates; at 14B, the penalty shrinks to a non-significant -4.6 pp. This shifts the aggregate from $+2.1$ pp (NS) to $+6.1$ pp (significant), driven by the interaction between yes/no penalty reduction and bridge advantage growth.

4.4 A Working Memory Hypothesis

We propose a *working memory hypothesis* to unify the pattern across scales and task types. Define the *working memory load* (WML) of a question as the number of distinct entity references that must be simultaneously tracked to produce the correct answer. Bridge questions typically require tracking 3–4 entities across 2–3 hops; yes/no questions require a single boolean judgment; LogicQA requires sequential constraint application against one “current state” at a time.

The hypothesis is:

- **WML exceeds model capacity:** A* externalizes bookkeeping that the model cannot hold internally. At 0.5B, multi-hop entity chains overflow this capacity; search helps.
- **WML within model capacity:** The branching overhead of A* adds noise without benefit. CoT’s single coherent trace is sufficient.
- **Formal logic:** WML is low and well-structured; search provides no benefit at any tested scale.

Important caveat. This is a hypothesis consistent with our data, not a demonstrated mechanism. We do not directly measure working memory capacity—we infer it from model scale and CoT performance. The hypothesis makes testable predictions: attention patterns across hop boundaries should differ between 0.5B and 8B on bridge questions, and the threshold scale at which search benefit disappears should correlate with the model’s multi-hop CoT accuracy. We propose these as directions for future mechanistic work.

4.5 Qualitative Analysis

We illustrate the mechanisms behind our quantitative findings with three representative examples drawn from the 8B HotpotQA matched evaluation set.

Example 1 — Bridge question: A* succeeds, CoT fails (premature commitment). **Question:** “When was the album that includes the song by Dustin Lynch released to country radio on February 17, 2017?”

Gold answer: September 8, 2017

CoT trace: “...the song was released on February 17, 2017. The answer is: February 17, 2017.”
(F1 = 0.25)

A* trace (final node):

Step 1: The song by Dustin Lynch released to country radio on February 17, 2017 is “Small Town Boy”.

Step 2: “Small Town Boy” appears on the album *Current Mood*.

Step 3: *Current Mood* was released on September 8, 2017. (F1 = 1.0)

CoT anchors on the February date in the question and reports it as the album release date. A*’s node-by-node expansion first resolves the song title, then the album, then the release date — keeping all three entities live in separate nodes rather than conflating them in a single linear trace.

Example 2 — A* overthinking: correct answer found early, then overwritten. **Question:** “Who directed the 2017 horror-thriller film in which Barry Keoghan, Nicole Kidman, and Colin Farrell starred?”

Gold answer: Yorgos Lanthimos

A* trace:

Step 1: “The Killing of a Sacred Deer is a 2017 psychological horror-thriller film directed by Yorgos Lanthimos. . . ” [h(n)=0 — goal reached]

Step 2: “The answer is: The Killing of a Sacred Deer” [OVERWRITES with film title instead of director name]

Final pred: “The Killing of a Sacred Deer” (F1 = 0.0)

The correct answer “Yorgos Lanthimos” is present at Step 1 (h=0). However the search budget is not exhausted, so expansion continues. Step 2’s goal-extraction prompt misidentifies the answer as the film title. This is the dominant A* failure mode (26% of bridge failures; see Appendix D).

Example 3 — Yes/No question: CoT succeeds, A* fails (format mismatch). **Question:** “Were Scott Derrickson and Ed Wood of the same nationality?”

Gold answer: yes

A* trace (final):

Step 1: “Scott Derrickson is an American director.”

Step 2: “Ed Wood was an American filmmaker.”

Step 3: “The answer is: The answer is: Yes” [double prefix — extraction fails]

Final pred: “The answer is: Yes” (F1 = 0.0)

CoT trace: “Both Scott Derrickson and Ed Wood are American filmmakers. The answer is: yes.” (F1 = 1.0)

A* generates a three-node search tree for a question that requires only a single comparison. The goal-extraction prompt produces a malformed output (double “The answer is:” prefix) because it is designed for span extraction, not boolean output. CoT handles the yes/no format naturally in one step.

5 Analysis

5.1 Heuristic Quality Validation

We validated the coverage heuristic on 500 manually annotated reasoning states (100 per dataset $\times$ 3 datasets, plus 200 additional HotpotQA). Human annotators scored reasoning progress on a 0–1 scale; the heuristic was computed automatically. The DeepSeek-14B critic used as $h(n)$ in the 14B experiments systematically underestimates human-assessed progress (mean 0.43 vs. 0.58, paired t -test $p < 0.001$, $\varepsilon_{\max} = 0.14$). This conservative-estimation property is consistent with the admissibility requirement and is stable across all three datasets.

5.2 Phase 2 Heuristic Ablation

To understand which component of $h(n)$ drives performance, we ran an ablation on StrategyQA with four configurations:

Coverage alone yields the best performance, +5.4 pp over BFS. Depth alone *hurts* (−1.4 pp). The combined configuration is unstable, suggesting the two components have conflicting incentives. We use coverage-only as the default heuristic.

Table 7: Heuristic ablation on StrategyQA (Llama-3.1-8B, 500Q).

Config	Heuristic	Accuracy (%)
A0	None (BFS, $h = 0$)	69.2
A1	Depth penalty only	67.8
A2	Coverage only	74.6
A3	Depth + coverage (combined)	unstable

Notably, BFS (no heuristic, $h = 0$) is equivalent to Tree-of-Thoughts-style undirected branching. BFS achieves 69.2% while full A* with coverage achieves 74.6%. The search structure matters, not just the branching.

An additional insight from inspecting the reasoning traces: **84.6% of correct A* answers on HotpotQA are found in exactly two reasoning steps** (step 1: identify the bridge entity; step 2: answer). The mean depth for correct answers is 2.08 steps vs. 1.95 for incorrect answers. This confirms that the typical successful A* trajectory is not a deep search—it is a two-hop entity chain that the model resolves in two steps with the coverage heuristic guiding it toward the right intermediate entity.

5.3 Oracle Gap and Method Complementarity

A practically important question is: how much accuracy is being left on the table by using only one method? For each dataset, we compute the *oracle accuracy*—the accuracy achievable if we always chose the better method per question—and the *complementarity*—the fraction of questions where one method succeeds and the other fails.

Table 8: Oracle gap and complementarity at Llama-3.1-8B and DeepSeek-14B. Complementarity = fraction of questions where the methods disagree (A*-only + CoT-only). Oracle = accuracy if the best method is always chosen.

Model	Dataset	A* (%)	CoT (%)	Oracle (%)	Gap to Oracle	Complementarity
Llama-8B	HotpotQA	70.1	68.0	81.4	11.3 pp	24.6%
	StrategyQA	72.6	71.2	83.0	10.4 pp	22.3%
	LogicQA	44.9	45.8	69.9	24.1 pp	49.2%
DeepSeek-14B	HotpotQA	82.5	76.4	86.4	3.9 pp	34.1%

Three observations stand out. **(1) LogicQA has the largest oracle gap (24.1 pp) and the highest complementarity (49.2%).** Despite neither method significantly outperforming the other in aggregate ($p=0.78$), the two methods succeed on largely *different* questions. Nearly half of LogicQA questions are solvable by exactly one method—pointing to substantial headroom for a learned routing policy. **(2) HotpotQA complementarity grows with model scale** (24.6% at 8B $\rightarrow$ 34.1% at 14B), suggesting that as both methods improve, they become more specialised rather than converging. **(3) The gap to oracle shrinks dramatically at 14B** (11.3 pp $\rightarrow$ 3.9 pp on HotpotQA), indicating that the 14B model is already near the routing ceiling on this task.

5.4 Question Complexity and A* Advantage

We examine whether question complexity—as measured by surface features—predicts A benefit within the HotpotQA bridge category (8B model, $N=567$).

Short questions (≤ 10 words) strongly favour CoT (-8.5 pp). These tend to be direct lookups or simple yes/no comparisons requiring no multi-hop chaining. As question length grows, the A advantage increases monotonically up to 16–20 words ($+6.8$ pp) before stabilising. This is consistent with the working memory hypothesis: longer questions typically encode more entity references and deeper hop chains, increasing the benefit of externalised bookkeeping.

Practical implication: Question length is a free, zero-cost signal available before any inference. A routing policy that uses only question word count would already achieve directional accuracy: choose CoT for questions under 10 words, A for questions over 15 words.

Table 9: A advantage by question length (8B HotpotQA, all questions, N=719). “Short” = ≤ 10 words; CoT consistently wins on short questions. A advantage grows with question length.

Question length	N	CoT (%)	A* (%)	Gap (pp)
Short (≤ 10 words)	118	74.6	66.1	−8.5
Medium (11–15 words)	262	67.9	70.2	+2.3
Long (16–20 words)	206	63.1	69.9	+6.8
Very long (> 20 words)	133	69.9	73.7	+3.8

5.5 Rule-Based Router: Implementation and Evaluation

The findings above — question type, question length, and model scale — define a simple rule-based router requiring *zero LLM calls and zero training*. We implement and evaluate it directly on our matched datasets.

Router implementation. The router applies rules in priority order:

1. If question type is yes/no (detected from first word or boolean pattern) $\rightarrow$ **CoT**
2. If question ≤ 10 words $\rightarrow$ **CoT**
3. If model ≤ 1 B parameters $\rightarrow$ **A***
4. If bridge question > 15 words $\rightarrow$ **A***
5. If question > 15 words $\rightarrow$ **A***
6. Otherwise $\rightarrow$ **A*** (default for multi-hop)

Question type is detected from surface patterns (first-word heuristic + boolean keyword matching); no NLP tools are required. Full implementation is provided in the supplementary code (`router.py`).

Evaluation. We evaluate the router on the 8B HotpotQA matched set (N=719), the dataset with the highest complementarity (24.6%).

Table 10: Rule-based router performance on 8B HotpotQA (N=719 matched questions). The router uses only question text — no LLM calls, no training data.

Method	Accuracy (%)	vs. Best Single
CoT alone	68.0	baseline
A* alone	70.1	+2.1 pp
Rule router	72.5	+2.4 pp
Oracle (upper bound)	81.4	+11.3 pp

The router achieves 72.5%, outperforming both A alone and CoT alone, and recovering 21.0% of the oracle gap. By routing yes/no questions (N=65) to CoT and bridge questions (N=567, especially those > 15 words) to A, the router captures the structural signal at zero inference cost.

Why rule-based rather than learned. We deliberately choose a rule-based design over a learned classifier for two reasons.

First, *data insufficiency*. The 177 discordant pairs available (questions where A and CoT disagree) are insufficient for robust supervised learning: 5-fold cross-validation with gradient boosting yields $57.2\% \pm 9.5\%$ accuracy on discordant pairs (barely above the 54.2% majority-class baseline), with high variance indicating overfitting on ~ 35 training examples per fold. Logistic regression fares similarly at $61.6\% \pm 3.3\%$.

Second, *topology features do not separate the discordant pairs at current scale*. We evaluated all 13 pre-generation structural features from our Reasoning Topology Framework (hop count, branching complexity, entity density, logical connective count, etc.) using the same cross-validation protocol. The best classifier achieves only $51.1\% \pm 3.2\%$ — essentially random. K-means clustering on the topology features produces four clusters, none of which is dominated

by A*-wins or CoT-wins; all clusters are dominated by “both correct” or “neither correct.” The 13 features characterise question *difficulty*, but do not reliably separate the winner within the difficulty band. This finding is itself informative: it means the topology features predict the existence of a multi-hop chain but not which strategy handles it better — that distinction requires observing both method outputs, which eliminates the pre-generation advantage.

The rule-based router, by contrast, directly instantiates the two statistically significant surface signals from this paper (question type at $p = 0.025$; question length at -8.5 pp): no training data, no overfitting, full transparency. The goal is to *explain* when search helps; a black-box learned classifier would answer “which method wins” without illuminating “why.”

The 79.0% remaining oracle gap motivates future work. A topology router trained on a *cross-dataset* corpus (combining HotpotQA, StrategyQA, LogicQA, and additional benchmarks) would have substantially more discordant examples and could plausibly recover a larger fraction. Section 4.4 outlines the mechanistic predictions that such a classifier should test.

5.6 Dual-Trace Synthesis

As an exploratory experiment, we investigated whether combining CoT and A* traces could outperform either alone. On LogicQA (100Q, Llama-3.1-8B): CoT alone achieves 39%, A* alone 33%, and a dual-trace synthesis (linearizing the A* trace and feeding both to the model for joint answer extraction) achieves 42%—a +3 pp improvement over the best single method. This confirms that the two strategies are complementary on at least some question types, and that the large oracle gap in Table 8 is not merely theoretical—the gain is partially reachable even without a trained router.

6 Discussion

When to use tree search in practice. Our results support the following decision table, derived directly from statistically significant findings:

Table 11: Practical decision guide derived from our findings. Each row is backed by a statistically significant result ($p < 0.05$).

Condition	Recommendation	Evidence
Question ≤ 10 words	Use CoT	-8.5 pp for A* (Table 9)
Question > 15 words (bridge)	Use A*	$+6.8$ pp for A* (Table 9)
Yes/No or boolean answer	Use CoT	-18.5 pp for A* at 8B ($p = 0.025$)
Multi-hop entity chain (bridge)	Use A*	$+7.5$ pp at 14B ($p < 0.001$)
Formal logic / MCQ constraints	Use CoT	Null effect at all scales
Model ≤ 1 B on multi-hop	Use A*	$+26.4$ pp at 0.5B ($p < 0.001$)

The single most reliable signal is question type: bridge/multi-hop questions benefit from A* across all three model scales tested, while yes/no and constraint-satisfaction questions consistently favour CoT. Question length provides a complementary zero-cost proxy when question type is not labelled.

Comparison with self-consistency and Tree-of-Thoughts. On Llama-3.1-8B, self-consistency (SC; majority vote over 5 samples) and Tree-of-Thoughts (ToT; BFS with LLM evaluation) achieve intermediate performance between CoT and A* on multi-hop tasks: StrategyQA (SC=71.2%, ToT=72.2% vs. A* =74.8%), HotpotQA (SC=78.2%, ToT=78.5% vs. A* =80.1%). On mathematical benchmarks, SC is competitive or slightly ahead (GSM8K: SC=86.9% vs. A* =86.1%) because sampling diversity suffices for well-structured sequential arithmetic. ToT collapses on math (GSM8K: 60.0%, MATH: 40.1%) due to evaluator failure on numeric outputs. This confirms that A*’s advantage on multi-hop QA comes specifically from *heuristic-guided* search—not generic branching (which ToT also provides) or sampling diversity (which SC provides). The coverage heuristic directs expansion toward unresolved entities, which is precisely the operation that distinguishes bridge-type multi-hop from sequential math.

Compute cost and the value of routing. A* uses approximately $24\times$ the tokens of CoT per question ($K = 3$ branches $\times$ average 5.9 nodes per search ≈ 18 LLM calls vs. 1 for CoT, each producing ~ 200 tokens). The rule-based

router’s practical value is avoiding this $24\times$ cost on the 152/719 questions (21%) where it routes to CoT—saving $\sim 70\%$ of A^* ’s total compute on those instances while maintaining higher accuracy than either fixed strategy. At scale, this cost reduction is substantial: for a deployment serving 10K questions/day, the router eliminates $\sim 2,100$ unnecessary A^* runs (each requiring 18 LLM calls), saving over 37,000 API calls daily.

Why aggregate numbers mislead. Aggregating across question types obscures the structural signal. On HotpotQA at 8B, A^* achieves +4.4 pp on bridge questions but -18.5 pp on yes/no questions, netting +2.1 pp overall. A practitioner who saw only the aggregate would dramatically underestimate both the benefit and the risk of using A^* . Task-type-aware evaluation is essential.

The non-monotonic scale curve. The HotpotQA curve—large at 0.5B, near-zero at 8B, larger again at 14B—is non-monotonic. One candidate explanation: at 0.5B search compensates for weak reasoning; at 8B the model internalizes multi-hop but produces weak heuristic estimates; at 14B the stronger model produces better heuristic estimates, making the search more directed on bridge questions. We note this explanation is not verified—we do not measure heuristic quality at 8B vs. 14B. An alternative is simply that the 8B overall result is noise ($p = 0.29$) and the true 8B bridge advantage (+4.5 pp, $p = 0.037$) is the stable signal, with 14B providing a stronger version of the same effect.

Limitations and open questions. We study three base model families; the working memory hypothesis is consistent with our data but not formally tested. The working memory hypothesis is post-hoc; attention-level analysis is needed to confirm the mechanism directly. The 27B experiments were incomplete due to infrastructure constraints. Future work should test: (a) instruction-tuned models, which may have different capacity profiles; (b) direct measurement of heuristic quality at each scale; (c) the working memory hypothesis via attention analysis.

7 Conclusion

We studied when A^* tree search improves over chain-of-thought prompting across three model scales and three task types. The statistically significant findings are: (1) a +26.4 pp advantage for A^* on multi-hop evidence synthesis at 0.5B scale ($p < 0.001$, $N=500$); (2) no significant overall advantage at 8B, but a significant +4.5 pp bridge-question advantage ($p = 0.037$) offset by a -18.5 pp yes/no disadvantage ($p = 0.025$); and (3) a significant +6.1 pp overall advantage at 14B ($p < 0.001$), driven by a +7.5 pp bridge advantage ($p < 0.001$). Formal logic shows no significant difference at any scale. We propose a working memory hypothesis—search helps when multi-hop entity-tracking demands exceed model capacity—as a unifying account of this pattern. The hypothesis is consistent with the data and makes testable mechanistic predictions, but is not yet directly verified. The most actionable takeaway is the question-type diagnostic: bridge questions reliably benefit from A^* across scales while yes/no and formal-constraint questions reliably do not, providing a practical routing criterion that does not require running both methods.

Limitations

- **Working memory not directly measured.** The hypothesis is post-hoc and consistent with the data; direct verification via attention analysis is future work.
- **8B overall results not significant.** None of the three 8B aggregate comparisons reach $p < 0.05$. The 8B contribution to the scale narrative rests on the question-type breakdown (bridge: $p = 0.037$, yes/no: $p = 0.025$), not the overall comparison.
- **27B incomplete.** Infrastructure failure prevented a fourth scale point on the curve.
- **Base models only.** Instruction-tuned and RLHF-fine-tuned variants may have different working memory profiles.
- **No token-cost analysis.** A^* uses $\sim 24\times$ CoT tokens ($K=3$, $B=15$). Accuracy-per-token tradeoff is not studied here.
- **Three datasets only.** Topology generalisation to math (GSM8K, MATH) and other multi-hop benchmarks (MuSiQue, 2WikiMultiHopQA) is left for future work.

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A Prompt Templates

A.1 HotpotQA CoT Prompt

You are a multi-hop reasoning assistant. Given a question and supporting context, reason step by step to find the answer.

Context: {context}

Question: {question}

Let's think step by step:

Step 1: [identify relevant entities from context]

Step 2: [chain the evidence]

...

The answer is: [final answer]

A.2 HotpotQA A* Node Expansion Prompt

You are building a reasoning chain step by step.

Context: {context}

Question: {question}

Current reasoning trace:

{trace_so_far}

Generate ONE next reasoning step. If you have enough information for a final answer, output:

"The answer is: [answer]"

Otherwise output the next intermediate fact.

CONFIDENCE: [0.0-1.0 how confident are you this step is correct]

A.3 StrategyQA CoT Prompt

Answer the following yes/no question by reasoning step by step.

At the end, output ONLY "yes" or "no" as the final answer.

Question: {question}

Answer: Let's think step by step.

Step 1: [first relevant fact or premise]

Step 2: [second fact or deduction]

...

Therefore, the answer is: yes / no

A.4 StrategyQA A* Node Expansion Prompt

You are building a reasoning chain step by step.

The final answer must be "yes" or "no".

Question: {question}

Current reasoning trace:

{trace_so_far}

Generate ONE next reasoning step. If you have gathered enough facts to answer, output:

"The answer is: yes" or "The answer is: no"

Otherwise, output the next intermediate fact.

CONFIDENCE: [0.0-1.0]

A.5 LogicQA CoT Prompt

Answer the following multiple-choice logic question.
Choose from options A, B, C, D.

Passage: {context}
Question: {question}
Options:
A. {option_a}
B. {option_b}
C. {option_c}
D. {option_d}

Let's reason step by step:
Step 1: [identify the key logical constraint]
Step 2: [apply the constraint]
...
The answer is: [A/B/C/D]

A.6 LogicQA A* Node Expansion Prompt

You are solving a multiple-choice logic problem step by step.
Final answer must be one of: A, B, C, D.

Passage: {context}
Question: {question}
Options: A. {a} B. {b} C. {c} D. {d}
Current reasoning trace:
{trace_so_far}

Generate ONE next reasoning step.
If you can now identify the correct option, output:
"The answer is: [A/B/C/D]"
Otherwise output the next logical deduction.

CONFIDENCE: [0.0-1.0]

B Reproducibility Details

Code and data will be released at <https://anonymous.4open.science/r/astar-reasoning-EC39>.

C Additional Results: Per-Dataset Raw Counts

C.1 Qwen-0.5B

HotpotQA CoT was re-run on the exact 500 questions used in the A experiment. The discordant pair breakdown is: A-only correct = 141, CoT-only correct = 9, confirming the amplification effect is highly asymmetric.

D Failure Taxonomy with Examples

We classified failures from two sources: (1) 43 bridge questions where A* failed and CoT succeeded (8B HotpotQA matched set), and (2) 50 bridge questions where CoT failed and A* succeeded. Each failure was categorised by inspecting the reasoning trace.

Table 12: Experimental configuration.

Parameter	Value
Random seed	42
CoT temperature	0.1
A* branch temperatures	{0.15, 0.45, 0.75}
A* branching factor K	3
A* budget B	15 nodes
A* max depth	8
Confidence clamping	$c_t \in [0.5, 0.99]$
Heuristic weights	$w_{\text{depth}} = 0.3, w_{\text{coverage}} = 0.3$
Qwen-0.5B inference	Ollama local
Llama-3.1-8B inference	vLLM (batched)
DeepSeek-14B inference	vLLM (batched)
HotpotQA split	validation
StrategyQA split	test
LogicQA split	test

Table 13: Qwen-0.5B accuracy on matched question sets (N=500 for all datasets).

Dataset	N	CoT	A*	Gap	p
HotpotQA	500	3.6%	30.0%	+26.4 pp	<0.001
LogicQA	500	24.0%	20.2%	−3.8 pp	NS
StrategyQA	110	56.4%	59.1%	+2.7 pp	NS

D.1 Failure Mode Distribution

Table 14: Failure mode distribution (bridge questions, Llama-3.1-8B HotpotQA, N=43 A* failures, N=50 CoT failures).

Failure Mode	A* failures	CoT failures
Overthinking (gold in trace, wrong final pred)	26%	—
Format mismatch (malformed extraction)	19%	—
Spurious chaining (wrong intermediate entity)	28%	—
Other / ambiguous	28%	—
Premature commitment (wrong entity at hop 1)	—	36%
Over-specificity (answer too long/specific)	—	32%
Missing entity hop (stopped one hop short)	—	10%
Other / ambiguous	—	22%

D.2 A* Failure Mode 1: Overthinking (26%)

Definition: The search finds the correct answer at an intermediate node ($h(n) = 0$) but continues expanding within the remaining budget. A later node overwrites the correct answer.

Real example:

Question: “Who directed the 2017 horror-thriller film in which Barry Keoghan, Nicole Kidman, and Colin Farrell starred?”

Gold: Yorgos Lanthimos

Node 1: “*The Killing of a Sacred Deer is a 2017 psychological horror-thriller film directed by Yorgos Lanthimos...*” [$h = 0$, goal state]

Node 2: “*The answer is: The Killing of a Sacred Deer*” [overwrites with film title]

Final pred: “The Killing of a Sacred Deer” (F1 = 0.0)

Root cause: Goal detection relies on terminal string patterns (“The answer is:”). When a non-terminal node contains such a pattern in its context, the search does not stop, and a later node produces a malformed answer. An LLM-as-judge goal detector would be more robust.

D.3 A* Failure Mode 2: Spurious Chaining (28%)

Definition: A* connects entities via an incorrect intermediate hop, producing a plausible but wrong answer.

Real example:

Question: “What was Iqbal F. Qadir on when he participated in an attack on a radar station located on the western shore of the Okhamandal Peninsula?”

Gold: flotilla

Node 1: “*Iqbal F. Qadir participated in an attack on a radar station...*” [$h = 0.5$]

Node 2: “*The answer is: A ship*” [partially correct entity type, wrong specificity]

Final pred: “A ship” (F1 = 0.0)

Root cause: The model correctly identifies the domain (naval vessel) but fails to narrow to the specific named entity (“flotilla”). The heuristic does not penalise over-general answers.

D.4 A* Failure Mode 3: Format Mismatch on Yes/No (19%)

Definition: A*’s node-expansion prompt is designed for span extraction. Boolean questions cause formatting failures in the goal-extraction step.

Real example:

Question: “Were Scott Derrickson and Ed Wood of the same nationality?” *Gold:* yes

Node 3: “*The answer is: The answer is: Yes*” [double prefix, extraction yields wrong string]

Final pred: “The answer is: Yes” (F1 = 0.0, CoT achieves F1 = 1.0)

Root cause: The node expansion prompt ends with “The answer is: [answer]”, which interacts with the model’s tendency to echo the template. A dataset-specific suffix (“yes or no:” for boolean questions) would resolve this.

D.5 CoT Failure Mode 1: Premature Commitment (36%)

Definition: CoT anchors on the first plausible entity it encounters and cannot revise.

Real example:

Question: “When was the album that includes the song by Dustin Lynch released to country radio on February 17, 2017?” *Gold:* September 8, 2017

CoT: “The song was released on February 17, 2017. **The answer is: February 17, 2017.**” (F1 = 0.25)

A*: Resolves song → album → album release date = September 8, 2017 (F1 = 1.0)

CoT conflates the song’s radio release date (given in the question) with the album’s release date (the answer). A*’s intermediate hops force explicit entity disambiguation.

D.6 CoT Failure Mode 2: Over-Specificity (32%)

Definition: CoT produces an answer that is technically correct but more specific than the gold span, failing the F1 threshold.

Real example:

Question: “What government position was held by the woman who portrayed the wife of the main character in the 1995 film Braveheart?” *Gold:* Chief of Protocol

CoT pred: “United States ambassador to Ghana and to Czechoslovakia, and also served as Chief of Protocol of the United States” (F1 = 0.0)

The answer contains “Chief of Protocol” but the surrounding text causes the F1 threshold to fail because token overlap is diluted. This is a metric artefact rather than a reasoning failure.